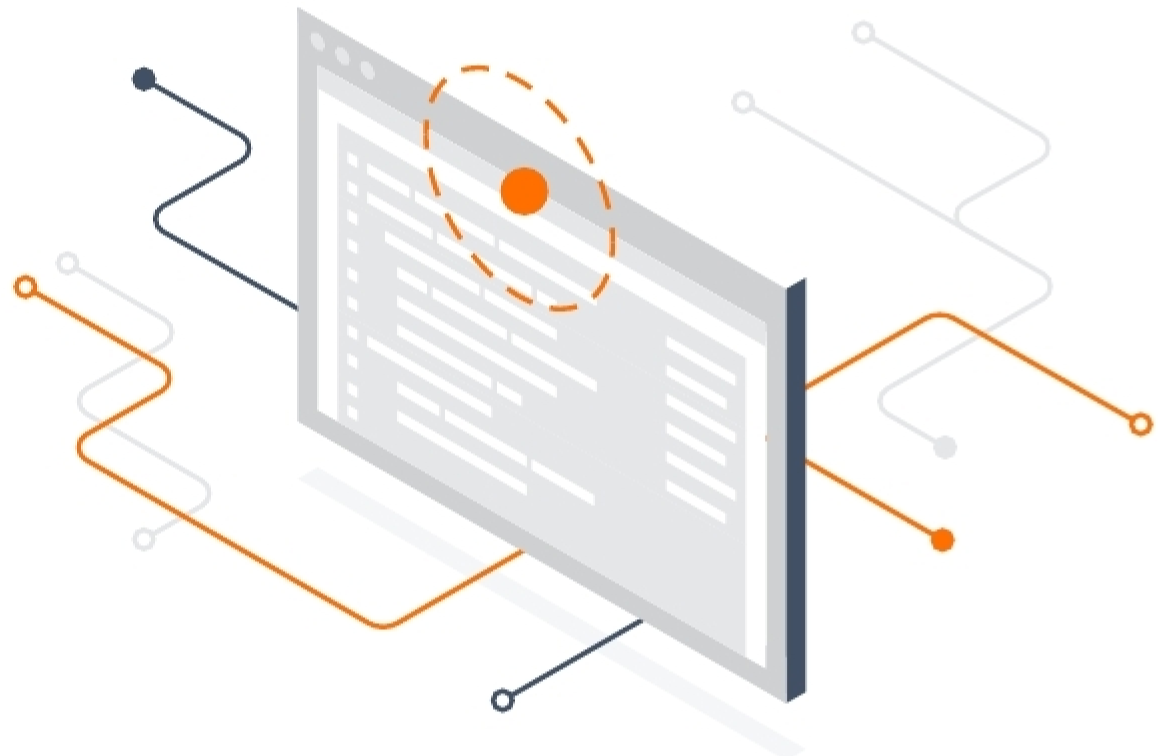
An Overview of TensorFlow.js

TensorFlow.js is a library for developing and training machine learning models in JavaScript and deploying them in a browser or on Node.js. It is an open source, hardware-accelerated JavaScript library for training and deploying machine learning models.



In recent times, a lot of attention is being paid to artificial intelligence (AI) and machine learning (ML). And in this context, the two most popular technologies are the Python and R environments or even C++ libraries. One of the most popular frameworks among developers is TensorFlow, which was developed by Google in 2011. Most of TensorFlow was designed in C++ and has bindings to Python or Java or R, but the most crucial language is missing, which is JavaScript.

In JavaScript, machine learning was performed on the server-side. A well-trained model was being deployed on the server and was available for access via the HTTP protocol. But in March 2018, the scenario changed and TensorFlow.js was launched by Google to perform ML and deep learning (DL) in JavaScript without having to use server-side applications. This was basically to define, train and run ML models in a Web browser, which were well equipped with a high-level layers API. TensorFlow.js supports WebGL and accelerates the code on the availability of the GPU. WebGL is the browser interface to OpenGL and enables the execution of JavaScript code on a GPU.

TensorFlow.js

TensorFlow.js especially allows users to build deep neural networks to integrate with existing or new Web apps. It has empowered a new set of developers from the extensive JavaScript community to enable new classes of on-device computation.

The interface of TensorFlow.js is strongly based on TensorFlow's high level API, Keras. The Keras code is often distinguishable from TensorFlow.js code only at second glance. Most differences are due to the different language constructs of Python and JavaScript for configuration parameters.

TensorFlow.js allows you to build ML projects from scratch. If the necessary data is available, models can be trained and executed directly in the browser. For this, TensorFlow.js uses the graphics card of the computer via the WebGL browser API. It does lose some performance as a result, because WebGL needs a few tricks to force it to execute the matrix multiplications required by TensorFlow.js. Yet these are necessary because TensorFlow.js, as a ML strategy, primarily supports neural networks. These networks can be mapped very well by matrix multiplications during training as well as during prediction.

Features of TensorFlow.js

- *Develops ML in the browser:* Uses flexible and intuitive APIs to build models from scratch with the low-level JavaScript linear algebra library or the high-level layers API.
- *Develops ML in Node.js:* Executes native TensorFlow with the same TensorFlow.js API under the Node.js runtime.
- *Runs existing models:* Uses TensorFlow.js model converters